1(a)(i)	direction or rate of transfer of (thermal) energy **or** (if different,) not in thermal equilibrium/energy is transferred	**B1**
1(a)(ii)	uses a property (of a substance) that changes with temperature	**B1**
1(b)	• temperature scale assumes linear change of property with temperature • physical properties may not vary linearly with temperature • agrees only at fixed points *Any 2 points.*	**B2**
1(c)(i)	$Pt = mc(\Delta)\theta$	**C1**
	$95 \times 6 \times 60 = 0.670 \times 910 \times \Delta\theta$	**M1**
	$\Delta\theta = 56\,°C$ so final temperature = 56 + 24 = 80 °C	**A1**
	or	
	$95 \times 6 \times 60 = 0.67 \times 910 \times (\theta - 24)$	**(M1)**
	so final temperature or $\theta = 80\,°C$	**(A1)**

Question	Answer	Marks
1(c)(ii)	**1.** sketch: straight line from (0,24) to (6,80)	**B1**
	2. temperature drop due to energy loss = (80 – 64) = 16 °C	**C1**
	energy loss = $0.670 \times 910 \times (80 - 64) = 9800$ J	**A1**
	or	
	energy to raise temperature to 64 °C = $0.670 \times 910 \times (64 - 24)$	**(C1)**
	= 24400 J	**(A1)**
	loss = $(95 \times 6 \times 60) - 24400 = 9800$ J	

Question	Answer	Marks